Training accuracy of every model after training to (nearly) 100 % training accuracy (TRAIN100), sorted from highest to lowest within each group.

| Model | Epochs (median) | Training accuracy |
|---|---|---|
| **(a) Main models, development cross-validation** | | |
| LightGBM on engineered features (LGBM-FEAT) | 2,000 trees | 1.000 |
| ECGFounder linear head (ECGF-linear) | 100 iter. | 1.000 |
| ECGFounder MLP head (ECGF-MLP) | 108 | 1.000 |
| Header + ECGFounder MLP fusion (FUSION-MLP) | 112 | 1.000 |
| C1: vote ensemble | 112[b] | 1.000 |
| IMG-vote: image-only vote | 108[c] | 1.000 |
| IMG-C3: image-only vote + prior rule | 108[c] | 1.000 |
| Image-only triage (AGT-img)[†] | 4,000 trees | 1.000 |
| MMCLEAN selective[†*] | n/r | 0.999 |
| C3: vote + nested prior rule | 112[b] | 0.995 |
| Image-only clinician features, logistic regression (CLIN-LR) | solver | 0.720 |
| Header + clinical features, logistic regression (HDRCLIN-LR) | solver | 0.589 |
| Header measurements, logistic regression (HDR-LR) | solver | 0.529 |
| **(b) Variants of every phase, development cross-validation** | | |
| E1 MLP, header + ECGFounder + feature families | 64 | 1.000 |
| E2 LightGBM, features + ECGFounder PCA | 2,000 trees | 1.000 |
| E7 mixup MLP | 144 | 1.000 |
| Multitask MLP (class + EF) | 131 | 1.000 |
| HIER two-stage | 108[c] | 1.000 |
| ENH M2 synthetic 10-s ECG → ECGFounder | solver | 1.000 |
| ENH M3 median-beat shape | solver | 1.000 |
| ENH M4 MiniRocket | solver | 1.000 |
| ENH M5 logistic regression | solver | 1.000 |
| ENH-vote | 108[c] | 1.000 |
| ULT M9 kNN | none | 1.000 |
| ULT M10 gradient boosting | 2,000 iter. | 1.000 |
| ULT M8 multi-scale ECGFounder LR | solver | 1.000 |
| E4 hierarchical MLP | 52 / 39 | 0.999 |
| E6 signal-quality gating | 112[b] | 0.988 |
| E3 weighted average of the C1 members | 112[b] | 0.985 |
| E3 stacking (logistic meta-learner) | 112[b] | 0.826 |
| ULT-stack | 108[c] | 0.794 |
| SUPER-TOP10-ORD, hospital members only | 108[c] | 0.783 |
| ULT M12 zone LR | solver | 0.624 |
| ENH M6 EF ridge → bands | exact | 0.557 |
| **(c) Label-quality studies (C3 recipe)** | | |
| HC30: reliable links only | 52[b] | 1.000 |
| NF: noise-free training | 45[b] | 1.000 |
| CLEAR: without border ECGs | 104[b] | 0.993 |
| VER: corrected labels | 132[b] | 0.986 |
| NFS: noise-free training, Severe kept | 44[b] | 0.985 |
| MMCLEAN: suspect Mild or Moderate labels dropped | 95[b] | 0.978 |
| **(d) Image and signal networks, original fixed split** | | |
| EfficientNet-B0 | 37 | 1.000 |
| HOG + RBF-SVM | solver | 1.000 |
| XGBoost (waveform features) | 2,000 trees | 1.000 |
| MobileNetV3-Large | 57 | 0.999 |
| 1-D SE-ResNet | 32 | 0.999 |
| ECGNet | 21 | 0.999 |
| ResNet-18 | 24 | 0.997 |
| ECGFounder fine-tuned (stages 5–6) | 12 | 0.996 |
| ConvNeXt-Atto | 53 | 0.996 |
| Header 0.6 + ResNet-18 0.4 | 24 (ResNet-18) | 0.972 |
| Logistic regression (header) | solver | 0.674 |
| **(e) Laptop run with EchoNext[*] (top-3 ensembles and their members)** | | |
| ECGF-lin | n/r | 1.000 |
| M5 | n/r | 1.000 |
| M8 | n/r | 1.000 |
| M9 | n/r | 1.000 |
| M10 | n/r | 1.000 |
| M12 | n/r | 1.000 |
| H-TAB-LR | n/r | 1.000 |
| H-TAB-GBM | n/r | 1.000 |
| H-X-GBM | n/r | 1.000 |
| H-MiMo-GBM (Mild vs Moderate)[‡] | n/r | 1.000 |
| EN-cls[°] | n/r | 1.000 |
| EN-gbm (H / plain)[°] | n/r | 1.000 |
| EN-gbm-X[°] | n/r | 1.000 |
| EN-MiMo-LR (Mild vs Moderate)[‡°] | n/r | 1.000 |
| EN-MiMo-GBM (Mild vs Moderate)[‡°] | n/r | 1.000 |
| EN-cls-all (H / plain)[°] | n/r | 0.999 |
| PIX-ZS (CNN on GPU)[°] | n/r | 0.999 |
| EN-mu-mid[°] | n/r | 0.999 |
| ECGF-MLP | n/r | 0.996 |
| EN-cls-all-X[°] | n/r | 0.996 |
| PIX-ZS-band[°] | n/r | 0.929 |
| EN-band-all (H / plain)[°] | n/r | 0.820 |
| SUPER (greedy) | n/r | 0.790 |
| EN-band[°] | n/r | 0.781 |
| SUPER-TOP10-ORD (best for 50s) | n/r | 0.748 |
| STAGED-4 (pool X) | n/r | 0.722 |
| M6 (EF bands) | n/r | 0.557 |
| EF-band | n/r | –[§] |

Epochs: median over the 15 folds; solver / trees / iter. = models not trained in epochs; [b] epochs of the neural member FUSION-MLP; [c] epochs of the neural member ECGF-MLP. [†] Selective model: training accuracy on the patients it answers. [‡] Two-class task (Mild vs Moderate). [°] Trained on EchoNext only (training accuracy on EchoNext). [*] Numbers from your other / laptop session, not re-run here; n/r = not reported. [§] Hidden by the scroll button in the screenshot. Logistic regressions on few inputs cannot reach 1.000; combiners whose decision rule is tuned on cross-validated member outputs (stacks, SUPER, STAGED) stay below 1.000 by design.